



Aror University of Art, Architecture, Design & Heritage Sukkur.

Department of Artificial Intelligence and Multimedia Gaming

Programming for AI (Spring-2026)

LAB No. 06

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Objective of Lab No. 06:

After performing lab 6, students will be able to:

- Handle runtime exceptions using try-except
- Use else and finally blocks
- Raise specific built-in exceptions
- Create and use custom exceptions
- Integrate exception handling with functions and logic

Lab Tasks:

1. Write a function `safe_division(a, b)` that:

- ☐ Returns the division result.
- ☐ Handles division by zero.
- ☐ Handles invalid input types.
- ☐ Always prints "Operation Completed", whether the exception occurs or not, it doesn't matter.

2. Create a function `validate_age(age)` that:



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- ☐ Raises ValueError if:
- ☐ Age is negative
- ☐ Age is greater than 120
- ☐ Otherwise prints "Valid Age"

3. Create a simple banking system.

- a. User has initial balance = 5000
- b. Write a function withdraw(amount)
- c. If amount > balance → raise custom exception InsufficientBalanceError
- d. If amount <= 0 → raise ValueError
- e. Otherwise deduct and print remaining balance

4. Create a small shopping cart system.

- a. Available products:

```
products = {  
    "laptop": 80000,  
    "mouse": 1500,  
    "keyboard": 3000  
}
```
- b. Function add_to_cart(product_name, quantity)
- c. If product not in dictionary → raise KeyError
- d. If quantity <= 0 → raise ValueError
- e. Create custom exception: OutOfStockError
- f. If quantity > 5 → raise OutOfStockError